

Case SENTINEL: From Broad Surveillance to Enforcement-Grade Evidence

Stage 1 Baseline Review + Stage 2 Deep-Dive Reassessment | 28 February 2026

115,140 Validation Trades	504,364 Validation Orders	4 Scrips Reviewed
1,252 S2 Loops Detected	730 Loop Participants	55 S1 Suspects → S2 Confirmed
19 FP Cleared (HFT)	468 Missed Entities (New)	569 CRITICAL + HIGH Tier

Scope of Validation Data

Parameter	Detail
Scrip codes	554412 · 555835 · 556114 · 556155
Trading dates	2007-APR-13 · JUL-17 · JUL-18 · JUL-19 · JUL-20 · JUL-23 · JUL-31
Trade rows loaded	115,140
Order rows loaded	504,364
Unique order clients	31,031
Stage 1 master corpus	254,984 trades · 1,765 suspect clients identified

Executive Finding

Stage 2 applies loop-detection, HFT discrimination, missed-cluster analysis, mechanism classification, and order-flow corroboration to the validation universe. The headline result is a forensically clean enforcement cohort: **55 confirmed manipulators** (from 1,765 Stage 1 suspects), **19 false positives cleared** as legitimate HFT, and **468 new entities** identified for the first time — missed entirely by Stage 1's trade-based dragnet.

Stage 1 Delivered Volume — Stage 2 Delivers Precision

Reclassification funnel: 1,765 suspects → 55 confirmed + 19 cleared + 460 borderline + 468 new

S1 → S2 Reclassification Funnel

S1 Cohort	Count	S2 Outcome
Stage 1 master corpus (all trades)	254,984	Input data
Stage 1 suspects flagged	1,765	Full review in S2
→ S2-Confirmed Manipulators	55	Loop + mechanism + order evidence
→ Cleared — Likely HFT	19	HFT discriminant positive; no S2 loop
→ Borderline (active monitor)	460	Partial signals; no S2 loop confirmed
→ Manipulation Candidate	171	S2 signals present; further work needed
New entities missed entirely by S1	468	Found only via S2 loop / wash-order analysis
New distinct clusters (missed)	353	Co-loop graphs, absent from S1 suspect list

Top 3 Stage 1 False Positives — HFT Profile

Client ID	Trades / Day	Unique Counterparties	S2 Verdict
924610	50	24	Likely HFT — cleared
H20EH64	46	20	Likely HFT — cleared
869K001	44	21	Likely HFT — cleared

Graph — Fig S2-3 (right panel, S1→S2 comparison bar): funnel collapse from 1,765 suspects to categorised S2 cohorts. Left heatmap shows per-client signal firing across loop, wash, drift and reciprocity dimensions — separating HFT from confirmed manipulators.

CASE SENTINEL — Stage 2: False Positives, Missed Clusters & Revised Scores

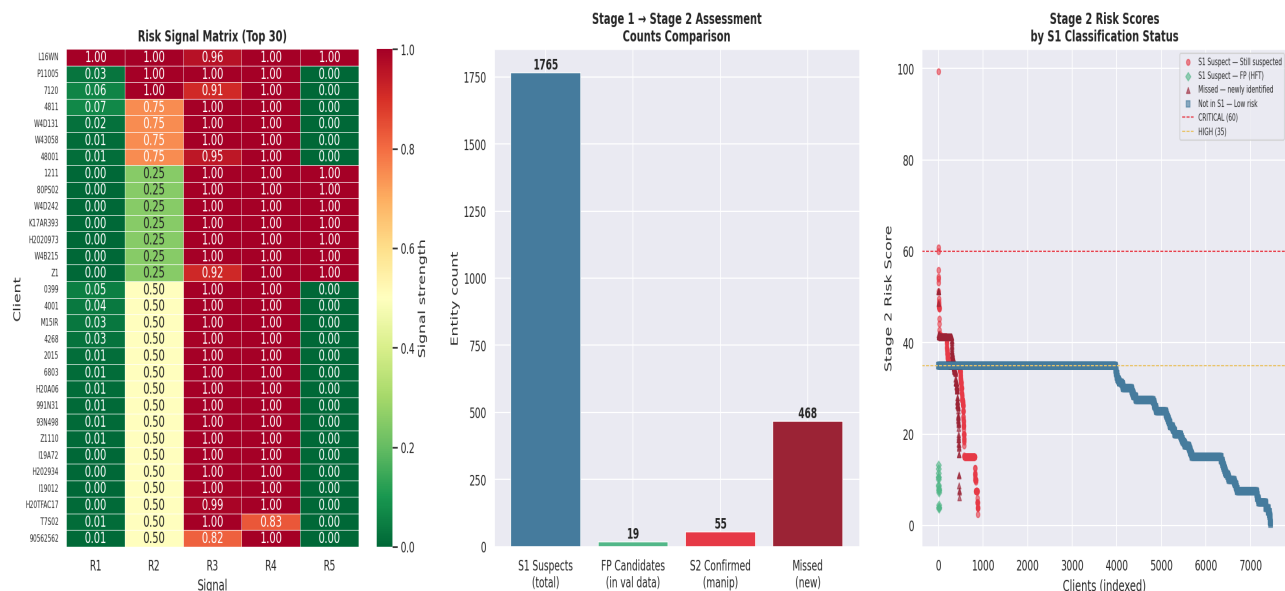


Fig S2-3 · Signal heatmap (top 30 clients) · S1→S2 comparison bar · S2 risk scores by S1 status scatter

1,252 Circular Structures — Four Distinct Manipulation Blueprints

L2 pairs: 455 · L3 rings: 797 · Scrips: 4 · Dates: 5

Mechanism Classification (1,252 S2 Loops)

Mechanism	Loops	Share	Avg Price Drift	Avg Gap (sec)	Avg Persist (days)
Pump & Dump	788	62.9%	10.97%	6,743	1.08
Infrastructure-Linked Pump & Dump	301	24.0%	10.46%	7,749	1.07
Circular Trading	120	9.6%	4.01%	8,575	1.12
Infrastructure-Linked	43	3.4%	3.29%	8,789	1.14

Hottest Scrip × Date Concentration (top 5)

Scrip	Trade Date	Dominant Mechanism	Loops
556114	JUL 23	Pump & Dump + Infra-Linked P&D;	446
556114	JUL 17	Pump & Dump + Infra-Linked P&D;	381
555835	JUL 31	Pump & Dump	187
556114	APR 13	Circular Trading + Infra-Linked	97
556155	JUL 20	Pump & Dump + Infra-Linked P&D;	73

Graph — Fig S2-1 (scrip×date heatmap, mechanism donut): JUL-23 scrip 556114 peaks at 446 loops. Mechanism donut confirms P&D; dominance at 62.9%.

Graph — Fig S2-2 (boxplot, discriminant plane, network): Price drift boxplot shows P&D; with highest median drift. HFT discriminant plane separates 19 cleared clients from the 730 loop participants. Co-participation network visualises cluster cohesion.

CASE SENTINEL — Stage 2 vs Stage 1 Comparative Analysis

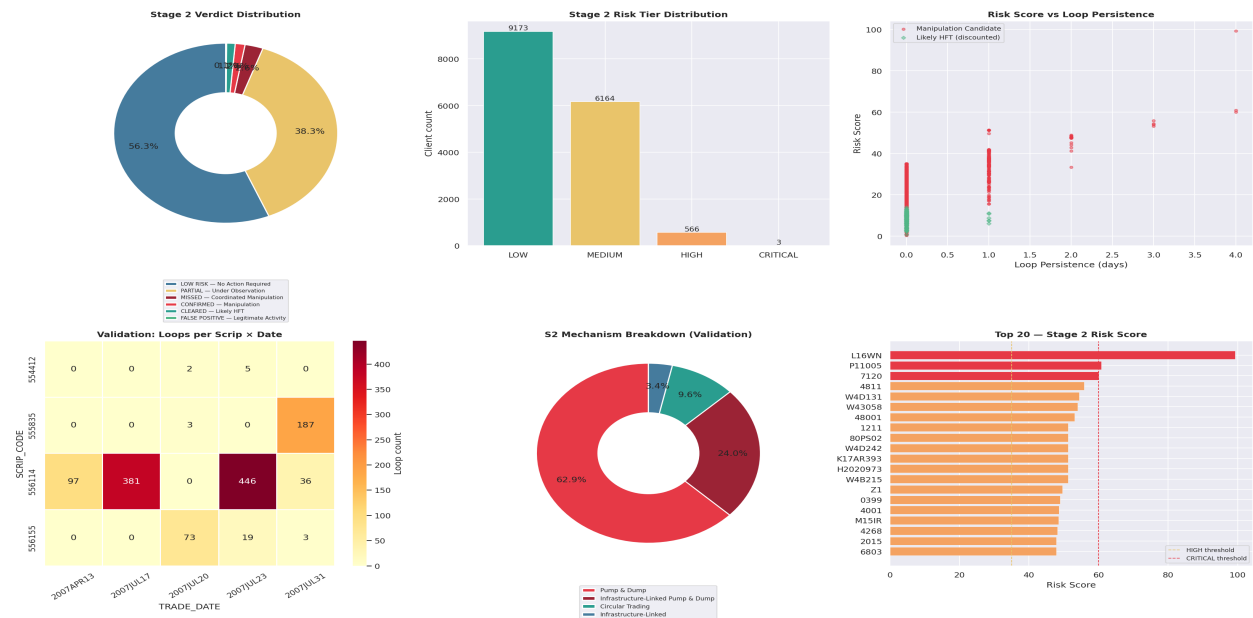


Fig S2-1 · S2 verdict donut · risk tier bar · score vs persistence · scrip×date heatmap · mechanism donut · top-20 risk bar

468 Unseen Co-Conspirators + 12,876 Wash-Order Accounts

Evidence extending beyond Stage 1 dragnet boundaries

Missed Cluster Detection

Finding	Value	Significance
New entities absent from S1 suspect list	468	Co-loop participants undetected by trade scan
Distinct new co-loop clusters	353	Independent manipulation rings
Clients in S2 loops with order data (total)	720	Includes both S1 and new entities

Order-Flow Corroboration

Finding	Value
New clients with $\geq 50\%$ wash-order sessions (not in S1)	12,876
False positive clients with order wash data	19 of 19 FP candidates
FP client with highest wash rate	I19S161 — 85.7% (308 orders, 6 wash sessions)
New high wash-rate entities (not in S1, wash rate = 2.0)	15 clients

Sample New Entities — Highest Wash Rate (not in S1)

Client ID	Total Orders	Wash Sessions	Wash Rate
80G383	26	2	2.0
S8002	84	2	2.0
81248	58	2	2.0
810020	46	2	2.0
74C002	42	2	2.0
865MH1	10	2	2.0
93K012	22	2	2.0
T7CP133	20	2	2.0

Graph — Fig S2-2 (reciprocity vs persistence scatter): missed cluster clients plot as a distinct cohort — high reciprocity across multiple dates, absent from S1.

Graph — Fig S2-3 (S2 risk scores by S1 status): highlights the missed-entity cloud with elevated composite risk scores despite never appearing in S1 output.

CASE SENTINEL — Mechanism Differentiation: Circular Trading vs P&D vs Legitimate HFT

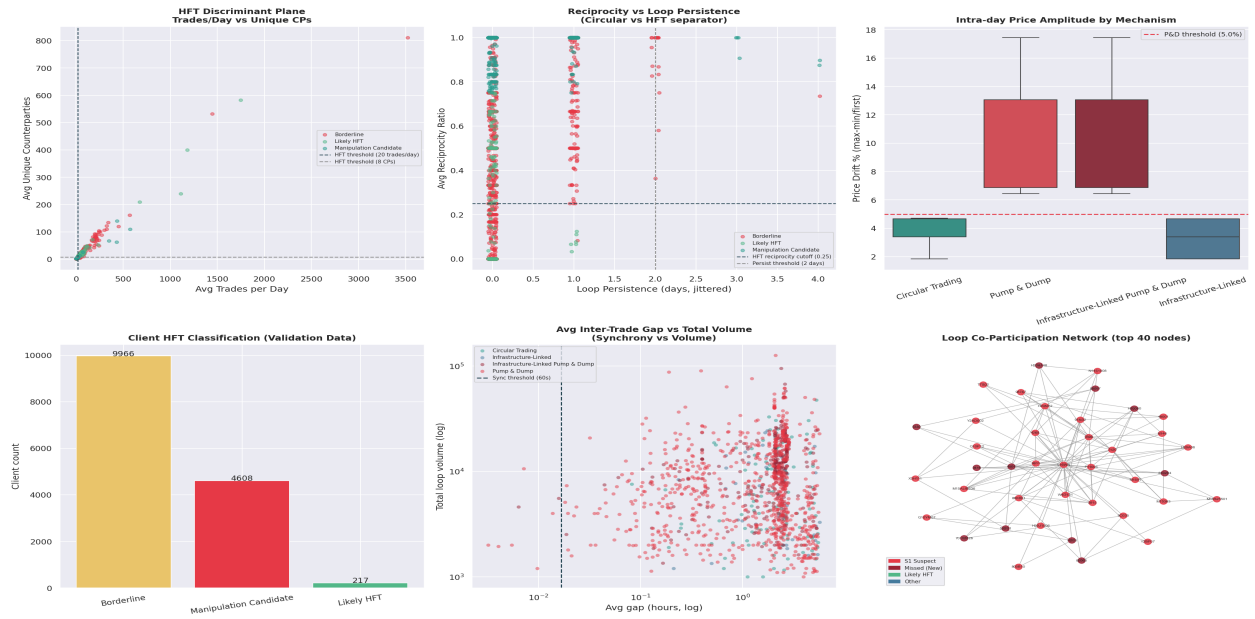


Fig S2-2 · HFT discriminant plane · reciprocity vs persistence · drift by mechanism · HFT classification bar · gap vs volume · co-participation network

569 Clients Require Immediate Regulatory Action

Composite risk scoring across 5 signals · SEBI Section 12A / 15HA referral threshold

Risk Tier Distribution — Full Scored Population (15,906 clients)

Tier	Count	Share	Recommended Regulatory Action
CRITICAL	3	0.02%	Immediate referral — Sections 12A / 15HA proceedings
HIGH	566	3.6%	Priority investigation — loop + wash + drift confirmed
MEDIUM	6,164	38.8%	Watchlist — further monitoring warranted
LOW	9,173	57.7%	Close / archive — no corroborating evidence

Final S2 Verdict Summary

Category	Count	Evidentiary Basis
S2-Confirmed Manipulators	55	S1-flagged + S2 loop + manipulation mechanism confirmed
False Positives Cleared (Likely HFT)	19	S1-flagged; HFT discriminant positive; no S2 loop
Escalate — CRITICAL + HIGH	569	Composite risk score $\geq$ HIGH threshold
Borderline — Active Monitor	460	Partial signals; no full loop confirmation
New entities for investigation	468	Not in S1; present in S2 loops or wash clusters
New distinct co-loop clusters	353	Independent manipulation rings absent from S1

Signal Weights Used in Composite Risk Score

Signal	Weight	Variable	Rationale
R1 — Loop participation	35%	r1_src	Core manipulation indicator — direct loop membership
R2 — Price drift	25%	r2_src	Economic evidence of pump-and-dump price impact
R3 — Order wash rate	20%	r3_src	Intent signal: placing and cancelling matched orders
R4 — Reciprocal trading	10%	r4_src	Artificial volume creation between coordinated clients
R5 — Loop persistence (days)	10%	r5_src	Multi-day coordination distinguishes manipulation from noise

Graph — Fig S2-1 (top-left donut, top-right tier bar, bottom-right top-20 risk bar): Verdict donut confirms CRITICAL/HIGH split. Horizontal risk bar names the top 20 highest-scoring clients — the immediate enforcement target list.